

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

John is organizing a fruit festival, and the quantities of various fruits are stored in a HashMap where fruit names are keys and quantities are values.

Help him develop a program to find the total quantity of fruits for the festival by summing up the values in the HashMap.

Input Format

The input consists of fruit quantities in the format 'fruitName:quantity', where fruitName is the name of the fruit(a string), and quantity is a double value representing the quantity.

The input is terminated by entering "done".

Output Format

The output prints a double value, representing the sum of values in the HashMap, rounded off to two decimal places.

If the value is not numeric, print "Invalid input".

If any special characters other than ':' are entered, print "Invalid format".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: Banana:15.2

Orange:56.3

Mango:47.3

done

Output: 118.80

Answer

```
import java.util.*;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        HashMap<String, Double> fruits = new HashMap<>();  
        double total = 0.0;  
        boolean invalidInput = false;  
        boolean invalidFormat = false;  
  
        while (true) {  
            String line = sc.nextLine().trim();  
            if (line.equalsIgnoreCase("done")) {  
                break;  
            }  
  
            // Must contain exactly one colon  
            if (!line.contains(":") || line.indexOf(':') != line.lastIndexOf(':')) {  
                invalidFormat = true;  
                break;  
            }  
        }  
    }  
}
```

```

String[] parts = line.split(":");
if (parts.length != 2) {
    invalidFormat = true;
    break;
}

String fruit = parts[0].trim();
String quantityStr = parts[1].trim();

// Validate fruit name: only alphanumeric (no spaces or symbols)
if (!fruit.matches("[A-Za-z0-9]+")) {
    invalidFormat = true;
    break;
}

// Validate numeric part
try {
    double quantity = Double.parseDouble(quantityStr);
    fruits.put(fruit, quantity);
} catch (NumberFormatException e) {
    invalidInput = true;
    break;
}

if (invalidInput) {
    System.out.println("Invalid input");
} else if (invalidFormat) {
    System.out.println("Invalid format");
} else {
    for (double qty : fruits.values()) {
        total += qty;
    }
    System.out.printf("%.2f", total);
}

sc.close();
}
}

```

Status : Correct

Marks : 10/10